

Practice Questions for 2024-10-01-SensitivityDuality

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1 Questions

1. In Slide 7, the optimal solution is given by $x_1^* = 25 - \frac{t}{40}$ and $x_2^* = 75 + \frac{t}{40}$. If we want to maintain feasibility ($x_1^*, x_2^* \geq 0$), what is the range of values for t ?
 - a. $-3000 \leq t \leq 1000$
 - b. $-750 \leq t \leq 1000$
 - c. $0 \leq t \leq 1000$
 - d. $-3000 \leq t \leq 0$
 - e. $-1000 \leq t \leq 3000$
2. In the context of Slide 1, if we increase the right-hand side constant of the second constraint ($10x_1 + 30x_2 \leq 4000$) by a small amount t , how does this affect the optimal values of x_1 and x_2 ?
 - a. x_1 increases and x_2 decreases proportionally.
 - b. x_1 decreases and x_2 increases proportionally.
 - c. Both x_1 and x_2 increase proportionally.
 - d. Both x_1 and x_2 decrease proportionally.
 - e. The effect on x_1 and x_2 is unpredictable without further information.
3. Based on Slides 2 and 3, how does the concept of complementary slackness help verify the optimality of the solution found in Slide 1?
 - a. Complementary slackness proves that the primal and dual objective functions are equal at the optimal solution.
 - b. Complementary slackness ensures that the constraints of both the primal and dual problems are satisfied.
 - c. Complementary slackness establishes a relationship between positive primal variables and zero dual slack variables (and vice versa), allowing for verification of the optimal solution.
 - d. Complementary slackness guarantees the existence of an optimal solution for both the primal and dual problems.
 - e. Complementary slackness is not directly relevant to verifying the optimality of the solution in Slide 1.

4. Referring to Slides 7-10, what is the economic interpretation of the dual variable y_2^* (the optimal dual variable associated with the second constraint in the primal problem) in the context of the linear program in Slide 1?
 - a. It represents the marginal cost of producing one unit of x_1 .
 - b. It represents the marginal cost of producing one unit of x_2 .
 - c. It represents the shadow price of the second resource (the resource constrained by the second constraint).
 - d. It represents the optimal value of the objective function.
 - e. It represents the marginal profit obtained by increasing the production of x_1 and x_2 .
5. In Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , what is the new optimal value of x_1 in terms of t ?
 - a. $25 + t/40$
 - b. $25 - t/40$
 - c. $75 + t/40$
 - d. $75 - t/40$
 - e. $25 + t$
6. According to Slide 8, what is the expression for the new optimal objective function value (ζ^*) after increasing the right-hand side of the second constraint in Slide 1 by t ?
 - a. $6250 - 0.75t$
 - b. $6250 + 0.75t$
 - c. $6250 + t$
 - d. $6250 - t$
 - e. $6250 + 32.5t$
7. Using complementary slackness (Slide 3), and given that $x_1^* = 25$ and $x_2^* = 75$, what is the value of y_1 ?
 - a. 0
 - b. 0.75
 - c. 32.5
 - d. 40
 - e. 70
8. What is the range of values for t that maintains feasibility ($x_1^*, x_2^* \geq 0$) in Slide 7, given the updated optimal solutions $x_1^* = 25 - t/40$ and $x_2^* = 75 + t/40$?
 - a. $t \geq 1000$
 - b. $-1000 \leq t \leq 1000$

- c. $-3000 \leq t \leq 1000$
 - d. $t \leq -3000$
 - e. $t \geq 3000$
9. In the context of sensitivity analysis for the linear program in Slide 1, how does the optimal dual solution (y^*) relate to the change in the optimal objective function value ($\Delta\zeta^*$) when there's a small change in the right-hand side vector (Δb)?
- a. The change in the objective function is independent of the dual solution.
 - b. $\Delta\zeta^*$ is equal to the dot product of Δb and y^* .
 - c. $\Delta\zeta^*$ is equal to the sum of the elements of y^* .
 - d. $\Delta\zeta^*$ is equal to the product of the largest element in Δb and the largest element in y^* .
 - e. The change in the objective function is inversely proportional to the magnitude of y^* .
10. In Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , what is the new optimal value of x_1 in terms of t ?
- a. $25 + t/40$
 - b. $25 - t/40$
 - c. $75 + t/40$
 - d. $75 - t/40$
 - e. $25 + t$

2 Answers

1. In Slide 7, the optimal solution is given by $x_1^* = 25 - \frac{t}{40}$ and $x_2^* = 75 + \frac{t}{40}$. If we want to maintain feasibility ($x_1^*, x_2^* \geq 0$), what is the range of values for t ?
 - a. The correct answer is A. To maintain feasibility, we need $25 - \frac{t}{40} \geq 0$ and $75 + \frac{t}{40} \geq 0$. Solving these inequalities gives $t \leq 1000$ and $t \geq -3000$, respectively. Therefore, the range of t is $-3000 \leq t \leq 1000$.
 - b. Incorrect. This range only considers the constraint $x_1^* \geq 0$.
 - c. Incorrect. This range only considers the constraint $x_1^* \geq 0$ and ignores the lower bound.
 - d. Incorrect. This range only considers the constraint $x_2^* \geq 0$ and ignores the upper bound.
 - e. Incorrect. This range has the bounds reversed.
2. In the context of Slide 1, if we increase the right-hand side constant of the second constraint ($10x_1 + 30x_2 \leq 4000$) by a small amount t , how does this affect the optimal values of x_1 and x_2 ?
 - a. This is incorrect. The changes in x_1 and x_2 are not simply proportional but are affected by the specific structure of the constraint matrix and the inverse of the basis matrix.
 - b. The analysis in Slide 7 shows that increasing the right-hand side of the second constraint by t leads to $x_1^* = 25 - \frac{t}{40}$ and $x_2^* = 75 + \frac{t}{40}$. Thus, x_1 decreases and x_2 increases as t increases.
 - c. This is incorrect. The changes in x_1 and x_2 are in opposite directions.
 - d. This is incorrect. The changes in x_1 and x_2 are in opposite directions.
 - e. This is incorrect. The effect can be determined from the sensitivity analysis presented in the slides.
3. Based on Slides 2 and 3, how does the concept of complementary slackness help verify the optimality of the solution found in Slide 1?
 - a. While complementary slackness is related to strong duality (the equality of primal and dual objective values at optimality), it doesn't directly prove it. Strong duality is verified by comparing the objective function values obtained from the primal and dual optimal solutions.
 - b. Complementary slackness doesn't directly ensure that all constraints are satisfied. Feasibility is checked separately by verifying that the constraints are satisfied by the optimal solution.
 - c. Slide 3 uses complementary slackness. Because $x_1^* > 0$ and $x_2^* > 0$, the corresponding dual slack variables z_1 and z_2 must be 0. This leads to a system of equations that can be solved to find the optimal dual variables, which are then used to verify strong duality.
 - d. Complementary slackness doesn't guarantee the existence of an optimal solution. The existence of an optimal solution is determined by the properties of the linear program itself (e.g., boundedness and feasibility).
 - e. Complementary slackness is a crucial tool for verifying optimality in linear programming.
4. Referring to Slides 7-10, what is the economic interpretation of the dual variable y_2^* (the optimal dual variable associated with the second constraint in the primal problem) in the context of the linear program in Slide 1?
 - a. This is incorrect. The marginal cost of producing x_1 is related to the objective function coefficient of x_1 , not the dual variable y_2^* .

- b. This is incorrect. Similar to option A, the marginal cost of producing x_2 is related to the objective function coefficient of x_2 , not the dual variable y_2^* .
 - c. Slides 7-10 show that the dual variables represent shadow prices. y_2^* represents the increase in the optimal objective function value for a one-unit increase in the right-hand side of the second constraint, which represents a resource. Therefore, it's the shadow price of that resource.
 - d. This is incorrect. The optimal value of the objective function is ζ^* , not y_2^* .
 - e. This is incorrect. The marginal profit is related to the changes in the objective function value, not directly to a single dual variable.
5. In Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , what is the new optimal value of x_1 in terms of t ?
- a. This incorrectly adds $t/40$ instead of subtracting it.
 - b. From Slide 7, the new optimal value of x_1 is given by $x_1^* = 25 - t/40$.
 - c. This is the new optimal value for x_2 , not x_1 .
 - d. This incorrectly subtracts $t/40$ from x_2 instead of x_1 .
 - e. This is an incorrect simplification of the change in x_1 .
6. According to Slide 8, what is the expression for the new optimal objective function value (ζ^*) after increasing the right-hand side of the second constraint in Slide 1 by t ?
- a. This incorrectly subtracts the change instead of adding it.
 - b. Slide 8 shows that the new optimal objective function value is $\zeta^* = 6250 + 0.75t$.
 - c. This is an oversimplification of the change in the objective function.
 - d. This is an oversimplification of the change in the objective function.
 - e. This uses the wrong dual variable as a multiplier for t .
7. Using complementary slackness (Slide 3), and given that $x_1^* = 25$ and $x_2^* = 75$, what is the value of y_1 ?
- a. This would imply that $x_1^* = 0$, which contradicts the given information.
 - b. This is the value of y_2 , not y_1 .
 - c. Since $x_1^* > 0$ and $x_2^* > 0$, we have $y_1 + 10y_2 = 40$ and $y_1 + 50y_2 = 70$. Solving this system gives $y_1 = 32.5$ and $y_2 = 0.75$.
 - d. This is the right-hand side of the first constraint in the dual problem.
 - e. This is the right-hand side of the second constraint in the dual problem.
8. What is the range of values for t that maintains feasibility ($x_1^*, x_2^* \geq 0$) in Slide 7, given the updated optimal solutions $x_1^* = 25 - t/40$ and $x_2^* = 75 + t/40$?
- a. This only considers the upper bound on t .
 - b. This is a narrower range than the one calculated in Slide 7.
 - c. Slide 7 explicitly states that to maintain feasibility, $-3000 \leq t \leq 1000$.

- d. This only considers the lower bound on t .
 - e. This is an incorrect range for t .
9. In the context of sensitivity analysis for the linear program in Slide 1, how does the optimal dual solution (y^*) relate to the change in the optimal objective function value ($\Delta\zeta^*$) when there's a small change in the right-hand side vector (Δb)?
- a. Answer A is incorrect because the optimal dual solution directly influences the change in the objective function value when the right-hand side changes, as shown by the formula in Slide 10.
 - b. Answer B is correct. Slide 10 explicitly shows that for small changes in the right-hand side, the change in the optimal objective function value is given by $\Delta\zeta^* = (\Delta b)^T y^*$, which is the dot product of the change in the right-hand side vector and the optimal dual solution vector. This demonstrates the crucial role of the dual variables (shadow prices) in sensitivity analysis.
 - c. Answer C is incorrect. The change in the objective function is not simply the sum of the dual variables; it's a weighted sum based on the changes in the right-hand side constraints.
 - d. Answer D is incorrect. The relationship between the change in the objective function and the dual solution is not based on the largest elements but on the dot product of the vectors.
 - e. Answer E is incorrect. The relationship is not an inverse proportionality; instead, it's a direct proportionality, as shown in Slide 10.
10. In Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , what is the new optimal value of x_1 in terms of t ?
- a. This is incorrect. The calculation in Slide 7 shows a decrease in x_1 as t increases.
 - b. From Slide 7, the new optimal value of x_1 after increasing the right-hand side of the second constraint by t is given by $x_1^* = 25 - t/40$.
 - c. This is incorrect. This represents the change in x_2 , not x_1 .
 - d. This is incorrect. This represents the change in x_2 with a negative sign, which is also incorrect.
 - e. This is incorrect. The change in x_1 is not a simple addition of t . The correct expression involves the inverse of the basis matrix.